

MASON JEWETT, MI, USA

WMO: 725417

Lat: 42.566N

Lon: 84.433W

Elev: 922

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.2	(c) 5.1	(d) -7.8	(e) 3.7	(f) 1.5	(g) -2.1	(h) 5.1	(i) 7.2	(j) 21.2	(k) 24.9	(l) 18.9	(m) 25.1	(n) 3.1	(o) 220	(p) 0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	89.6	72.2	86.3	70.9	83.6	69.5	75.3	83.9	73.6	82.1	72.0	80.0	7.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	124.8	80.0	71.5	120.3	78.9	69.8	113.5	77.1	39.3	84.5	37.8	81.8	36.3	79.6	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	17.1	15.3	DB	-8.9	93.6	6.0	5.4	-13.2	97.5	-16.7	100.7	-20.1	103.8	-24.4	107.7
			WB	-9.5	79.5	5.9	4.5	-13.7	82.7	-17.1	85.3	-20.4	87.9	-24.7	91.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	24.0	25.7	35.4	47.7	58.7	67.7	71.1	69.3	62.9	51.2	40.0	29.2		
	DBStd	18.96	11.24	10.87	11.06	9.26	8.76	6.68	5.78	5.52	7.68	8.92	9.11	9.91		
	HDD50	3198	808	682	472	151	21	0	0	0	3	95	320	646		
	HDD65	6557	1272	1100	918	524	235	47	14	22	130	437	749	1109		
	CDD50	2723	1	2	19	81	292	532	653	599	389	132	21	2		
	CDD65	608	0	0	1	4	41	129	202	155	66	10	0	0		
	CDH74	5319	0	0	10	49	400	1113	1855	1252	563	77	0	0		
	CDH80	1520	0	0	1	5	98	331	595	329	149	12	0	0		
	WSAvg	6.1	7.1	7.2	7.2	7.7	6.2	5.2	4.4	4.0	4.5	5.9	6.6	6.9		
	PrecAvg	34.6	2.3	2.0	2.4	3.0	4.0	3.9	3.5	2.9	3.1	2.9	2.7	2.0		
(15) Precipitation	PrecMax	41.9	4.2	3.7	4.2	6.1	9.6	7.4	6.6	6.2	9.8	6.6	5.2	3.7		
	PrecMin	27.0	0.7	0.8	0.7	0.4	1.9	2.0	0.9	0.5	0.7	0.4	1.3	0.8		
	PrecStd	4.1	1.1	0.9	0.9	1.3	2.0	1.4	1.3	1.5	2.0	1.4	1.2	0.7		
	PrecStd	4.1	1.1	0.9	0.9	1.3	2.0	1.4	1.3	1.5	2.0	1.4	1.2	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	58.8	74.6	80.5	87.6	90.4	92.8	91.1	89.6	81.7	70.0	58.8		
		MCWB	52.3	52.8	63.0	61.8	69.3	72.0	73.6	72.1	70.0	69.4	57.5	54.6		
	2%	DB	48.4	52.3	63.9	73.3	82.2	87.7	89.7	87.1	84.2	75.3	62.8	52.3		
		MCWB	44.9	48.3	54.7	58.1	66.2	71.0	73.6	71.7	69.4	63.0	53.5	48.9		
	5%	DB	43.2	45.1	57.4	69.8	79.0	83.9	86.3	83.5	80.7	70.7	58.6	46.1		
		MCWB	39.6	40.7	50.5	56.3	64.2	69.3	71.7	69.8	66.9	61.2	51.9	42.9		
	10%	DB	38.4	39.2	52.4	64.1	73.8	81.0	82.5	81.1	76.8	66.0	54.4	41.5		
		MCWB	36.1	35.7	46.2	53.2	61.8	67.5	69.6	68.8	65.4	58.8	48.4	38.4		
	0.4%	WB	53.7	54.8	64.0	65.3	72.2	75.3	78.3	76.9	74.7	70.8	59.7	55.2		
		MCDB	55.4	57.3	73.6	75.9	81.8	85.2	87.6	86.3	83.8	80.3	66.1	57.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.1	48.1	57.1	61.4	69.0	73.2	75.9	74.3	72.1	66.7	55.8	49.5		
		MCDB	47.2	51.6	62.1	68.2	78.5	83.0	84.7	82.0	79.6	72.2	61.2	52.4		
	5%	WB	40.0	41.1	52.4	58.3	67.0	71.4	74.2	72.6	70.0	62.9	52.7	43.4		
		MCDB	43.0	44.9	57.0	67.0	75.1	80.2	82.7	80.1	76.4	67.9	57.6	45.8		
	10%	WB	36.4	36.2	46.3	54.5	64.3	69.6	72.3	70.9	67.6	59.3	48.7	38.6		
		MCDB	38.6	39.5	51.0	63.0	71.7	77.4	80.2	77.9	74.6	65.2	53.5	41.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	14.5	16.8	20.0	19.9	19.2	20.5	19.8	20.6	18.0	14.6	11.6		
		MCDBR	16.2	18.5	24.2	26.3	23.8	22.3	23.0	22.2	23.9	23.9	20.9	16.7		
		MCWBR	13.8	15.4	16.5	14.8	12.2	10.0	10.2	10.1	11.1	13.6	14.6	14.2		
		MCDBR	14.9	18.0	22.5	22.8	20.4	18.7	18.9	17.7	18.8	20.0	19.2	16.4		
	5% WB	MCWBR	13.5	16.1	17.1	15.5	12.5	10.1	10.0	9.6	10.5	14.0	15.5	14.8		
		MCWBR	13.5	16.1	17.1	15.5	12.5	10.1	10.0	9.6	10.5	14.0	15.5	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.345	0.342	0.387	0.417	0.438	0.427	0.421	0.381	0.345	0.314	0.296		
	taud		2.370	2.190	2.359	2.253	2.220	2.197	2.235	2.247	2.345	2.443	2.477	2.414		
	Ebn at Noon		267	266	286	279	272	266	268	266	269	267	260	257		
	Edh at Noon		25	36	35	42	44	45	43	41	35	28	23	23		
(44) All-Sky Solar Radiation	RadAvg		464	746	1138	1449	1690	1909	1940	1664	1341	860	535	368		
	RadStd		42	95	115	133	155	120	90	76	125	83	69	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page